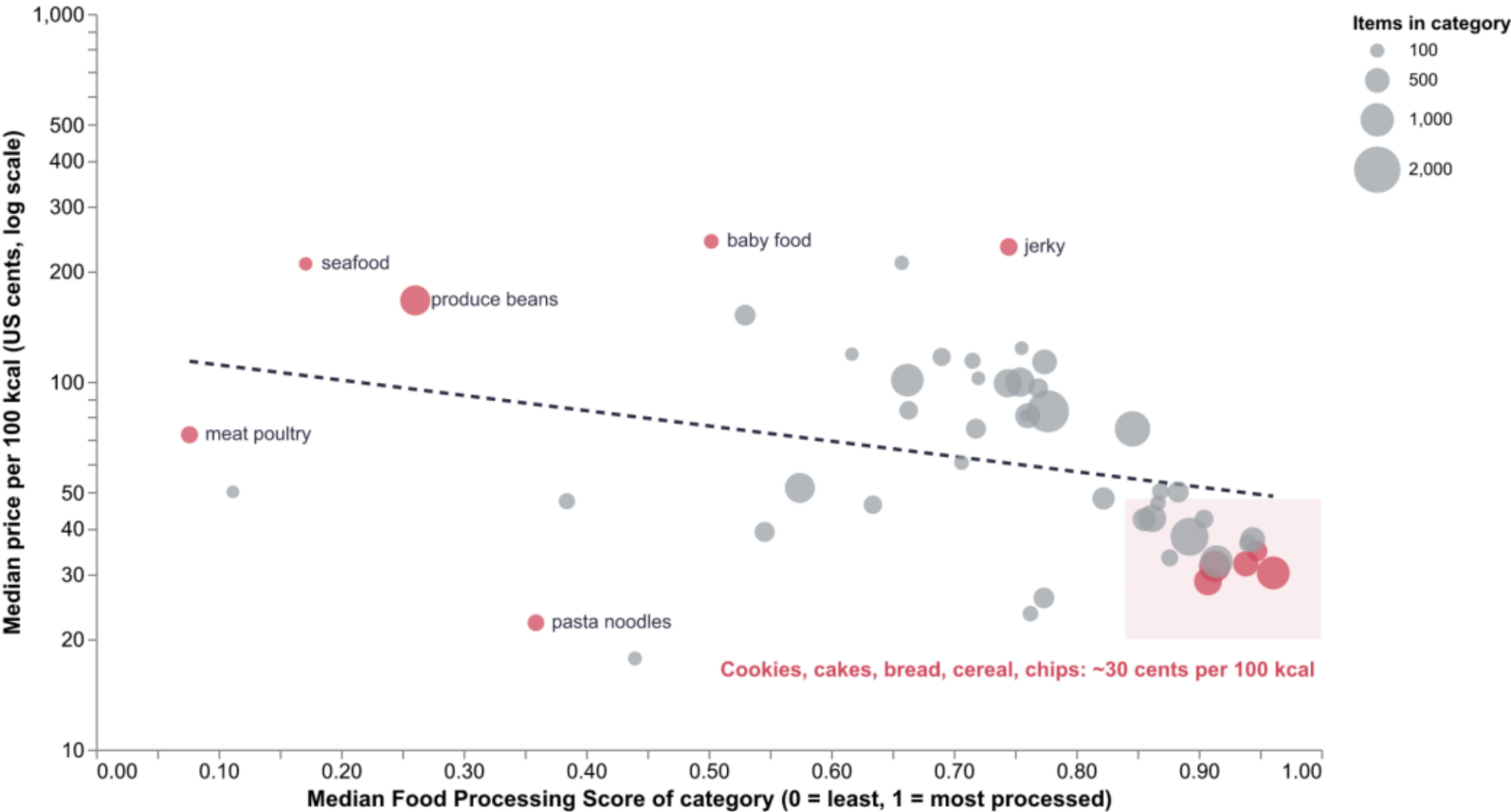


Plot A

The more processed the food, the cheaper its calories

Each dot is one of 50 food categories (categories with 20+ items priced per calorie; coffee beans excluded).
Value = category median across Walmart, Target and Whole Foods items. Dashed line: log-linear fit.

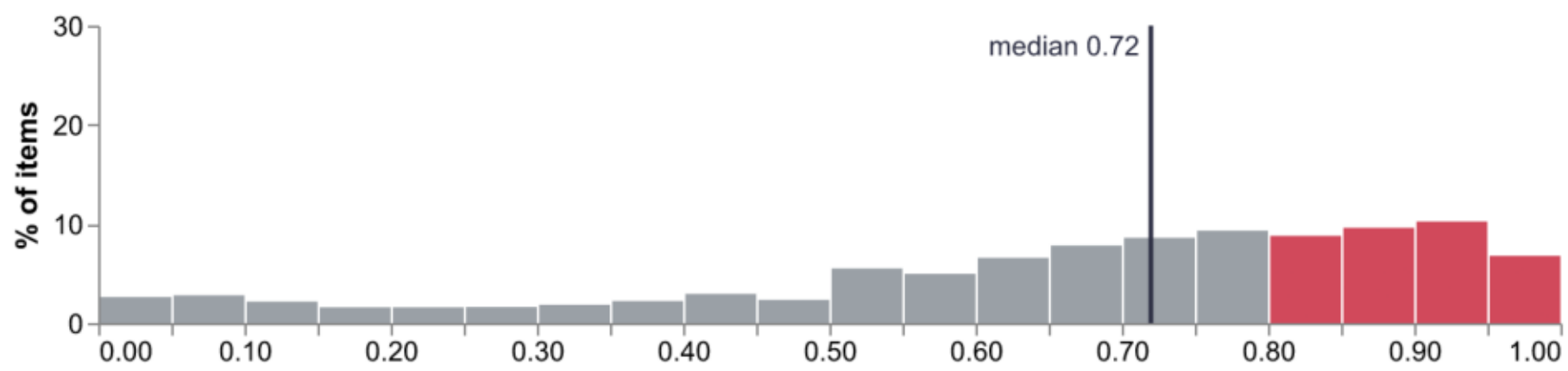


Plot B

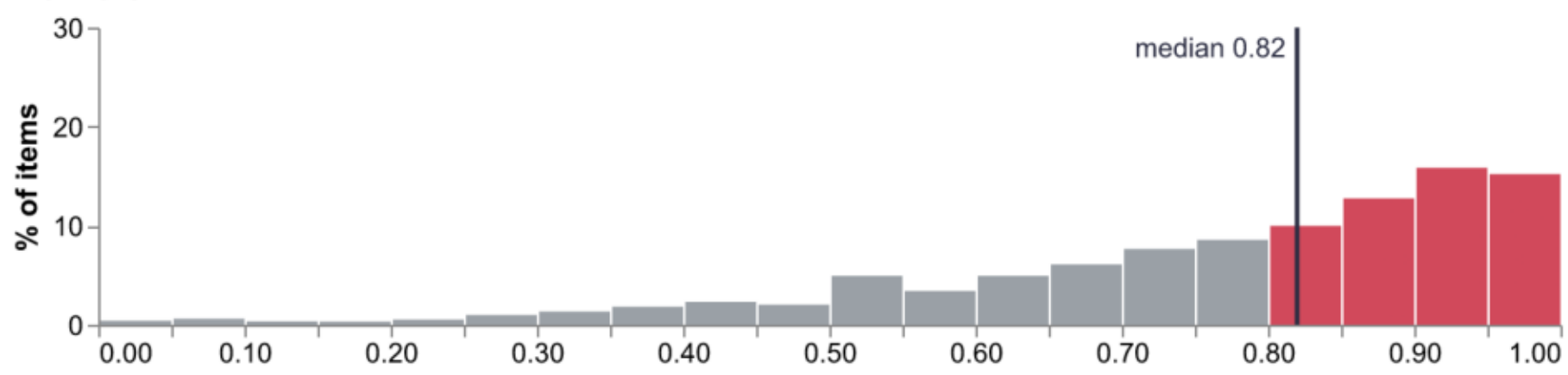
Whole Foods stocks far more minimally processed food than Walmart or Target

Histogram of Food Processing Score per store (20 bins, all 26,250 items); red = score 0.8 or higher (heavily processed).

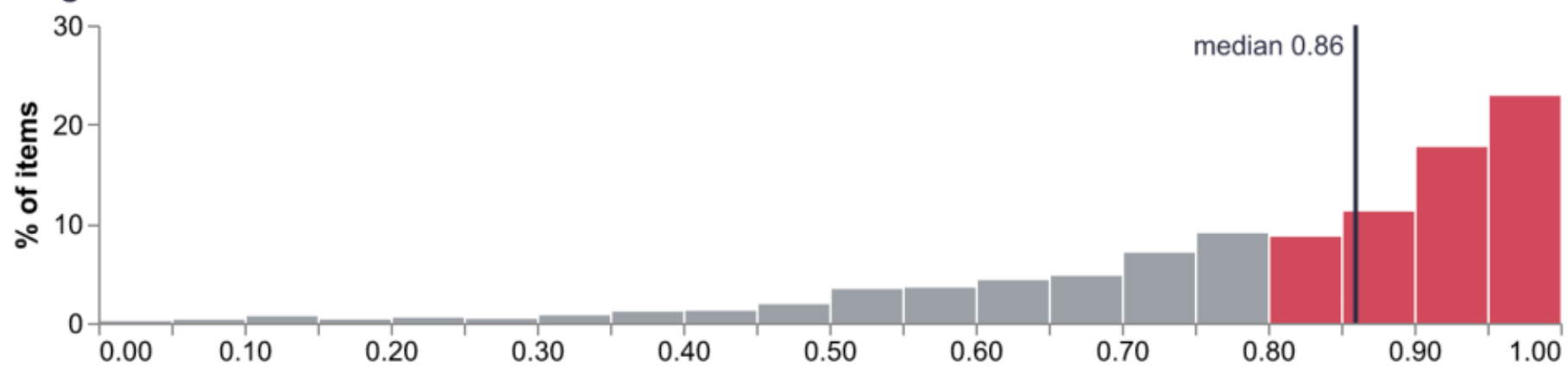
Whole Foods



Walmart



Target

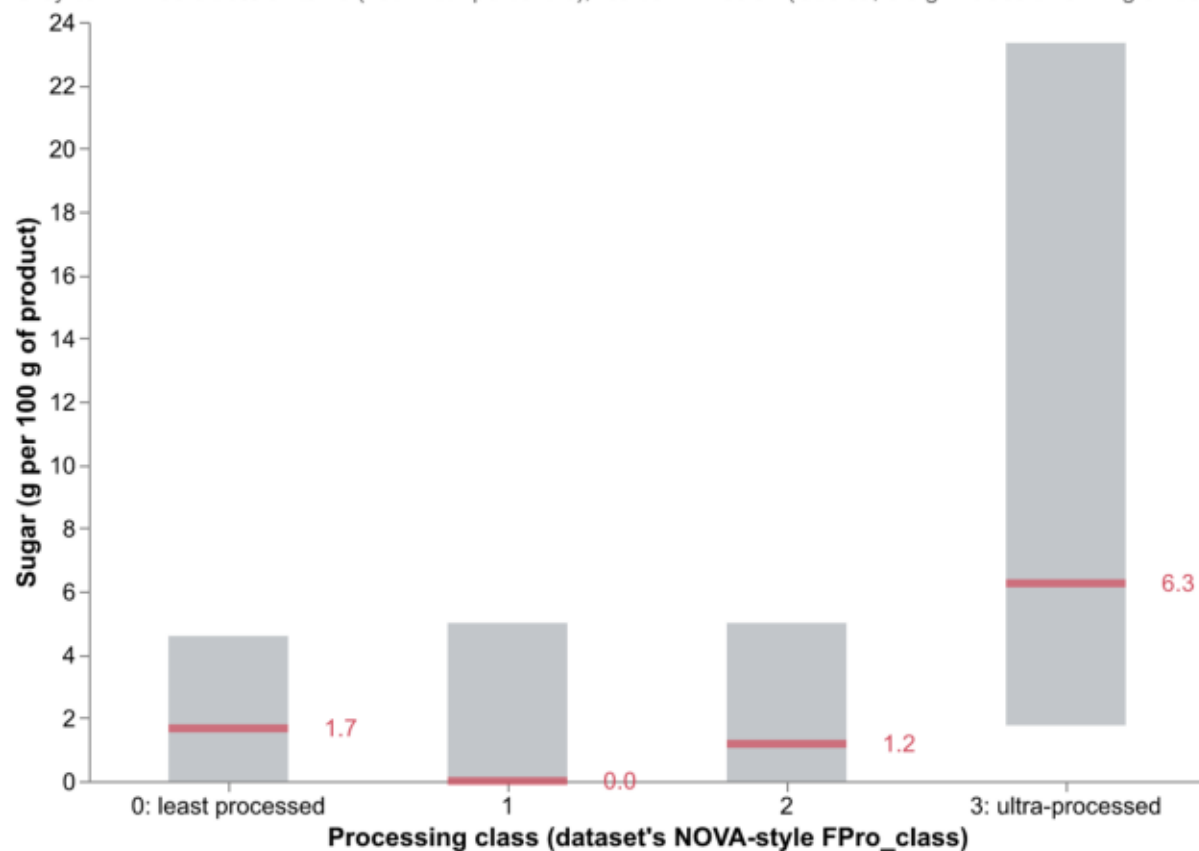


Food Processing Score (0 = least, 1 = most processed)

Plot C

Sugar jumps only in the ultra-processed class

Grey bar = middle 50% of items (25th-75th percentile); red tick = median (labeled; 6.3 g in class 3 vs 1.7 g or less elsewhere). Items with sugar > 100 g/100 g (data errors, 38 rows) removed.



Preferred plot

I prefer Plot A, the price-per-calorie scatter. It's the only one of the three that ties processing to something a shopper feels at the register, so it argues something instead of just describing a distribution. Both axes use position, the most accurate channel for comparing two quantities, and price is on a log scale because otherwise the cheap, heavily processed categories get squashed against the bottom. Plotting every product would be a smear, so each dot is a category median and its area shows how many items sit behind it. A 20-item category shouldn't look as solid as a huge one. Color only marks the categories I talk about, and the shaded box around cookies, cakes, bread, cereal and chips (about 30 cents per 100 kcal) is where I want people to look first. Plot B is easier to read, but part of what it shows is just what each store stocks, since Whole Foods has categories the others don't. Plot C is the weakest because sugar doesn't climb steadily across classes. Only class 3 stands out. Plot A has its own weakness: the trend is noisy (Spearman about -0.50 across categories) and a median hides the spread inside each category. In the final version I'll say that plainly and check whether item-level detail, or holding the store constant, makes the pattern clearer.